CSAI 202

Introduction to Database Systems



Lec 01

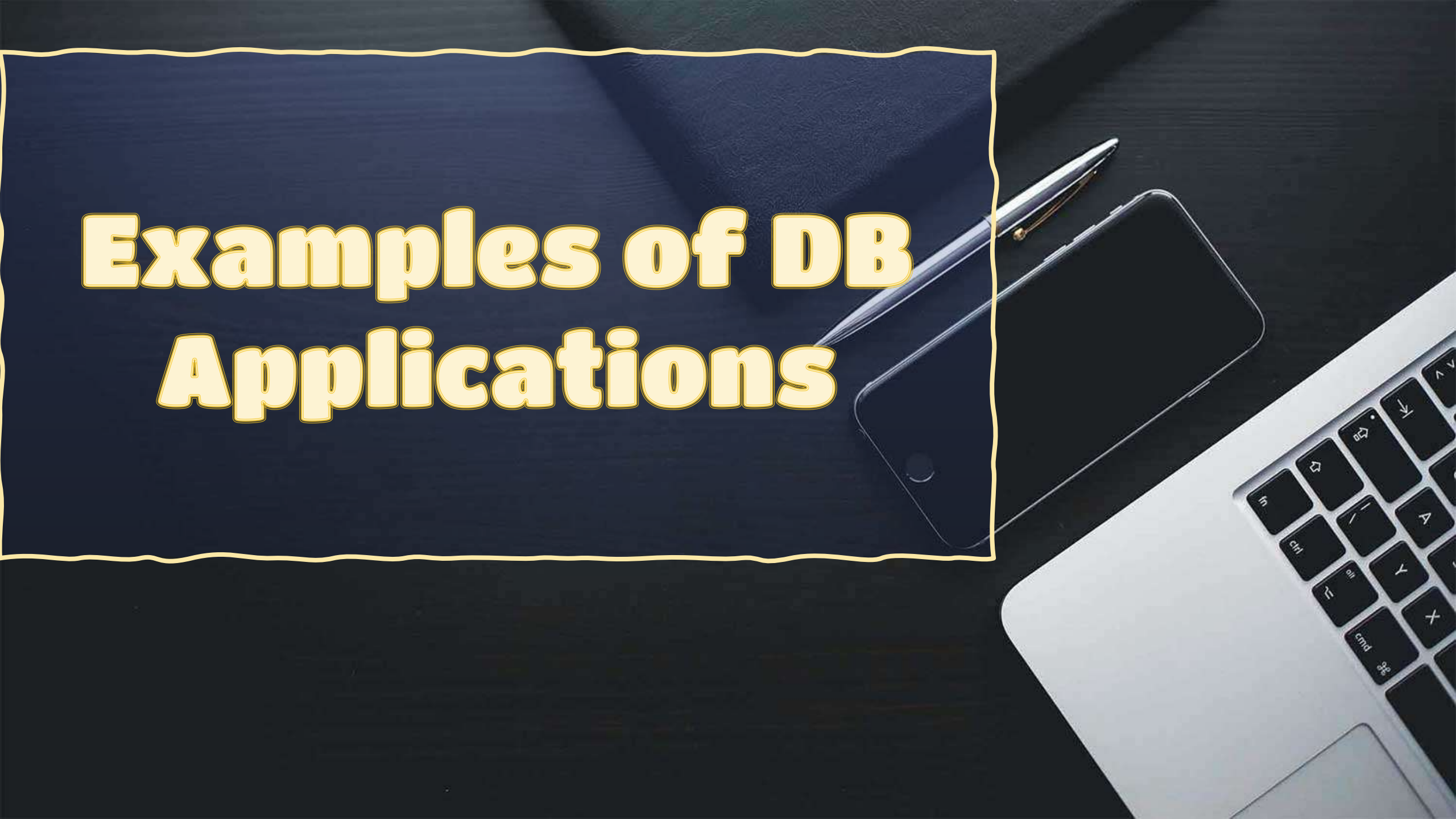
DB and DB Users



Outline

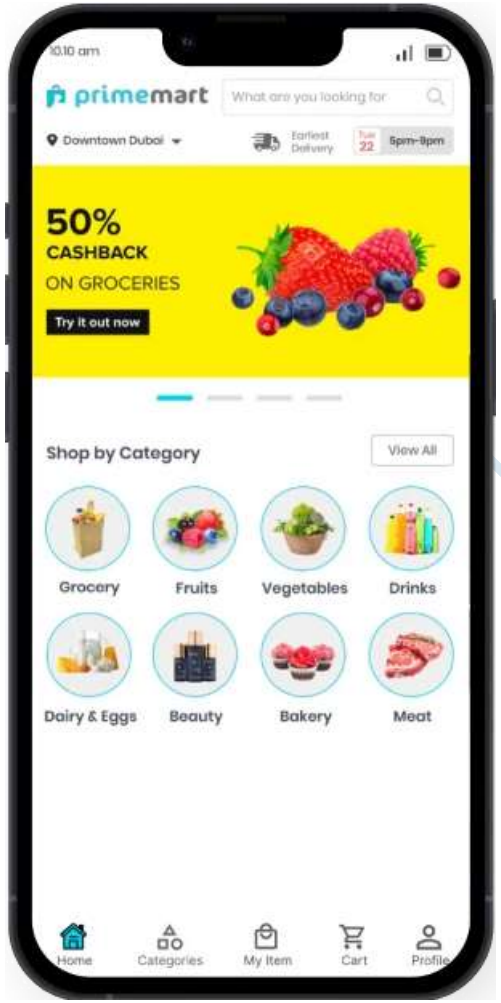
- Examples of Database Applications
- Basic Definitions
- Database System Environment
- Example of a Database (UNIVERSITY)
- Database vs. File Systems
- Main Characteristics of the Database Approach
- Typical DBMS Functionality
- When Not to Use Database?

Examples of DB Applications



Traditional Applications

- Purchases



- Reservations



Recent Applications

- Multimedia databases



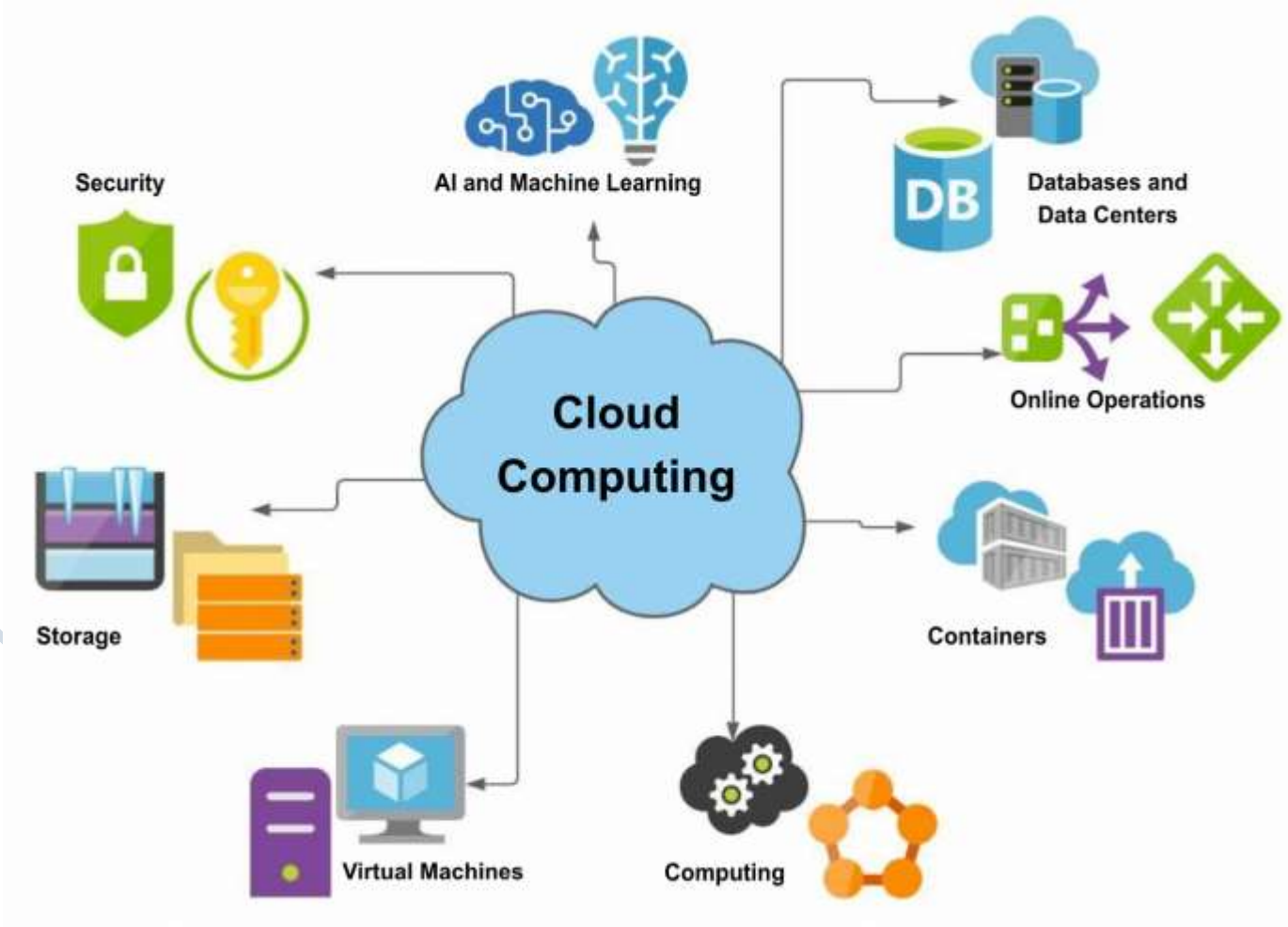
Recent Applications

- Social media databases (Facebook, Twitter,…..)



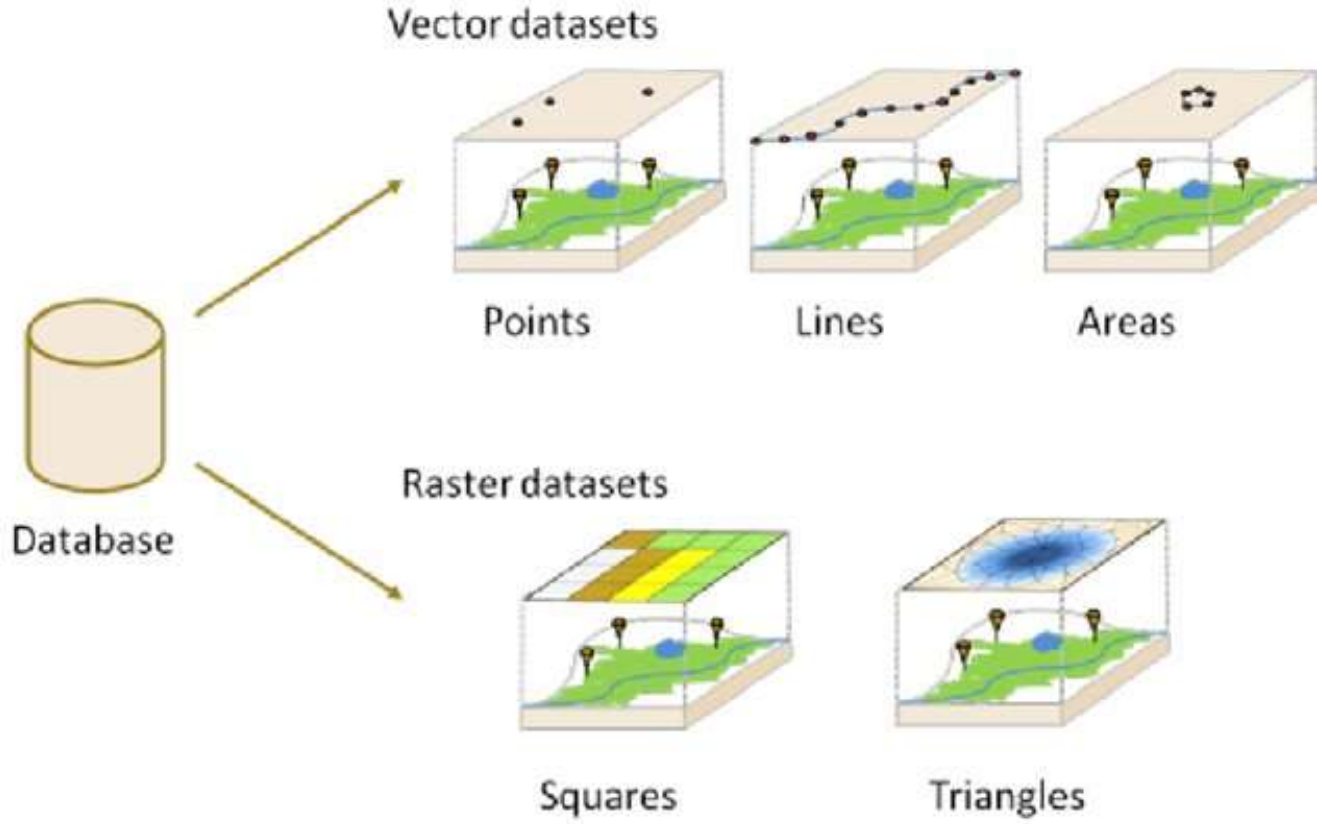
Recent Applications

- Mobile databases



Recent Applications

- Geographic Information Systems (GIS)



Basic Definitions

- Database:
  - A collection of related data.
- Data:
  - Known facts that can be recorded and have an implicit meaning.
- Mini-world:
  - Some part of the real world about which data is stored in a database. For example, student grades and transcripts at a university.

Basic Definitions

- Database Management System (DBMS):
  - ★ A software package/ system to facilitate the creation and maintenance of a computerized database
  - ★ A general-purpose software system that facilities the process of defining , constructing, manipulating, and sharing databases among different users and applications.

Typical DBMS Functionality

- **Define** a particular database in terms of its ***data types, structures***, and ***constraints***.
- **Construct** or Load the initial database contents on a ***secondary storage medium***.
- **Manipulating** the database:
  - ***Retrieval***: Querying, generating reports
  - ***Modification***: Insertions, deletions and updates to its content
- **Accessing** the database through Web ***applications***
- Processing and **Sharing** by a set of concurrent users and application programs – yet, keeping all data valid and consistent

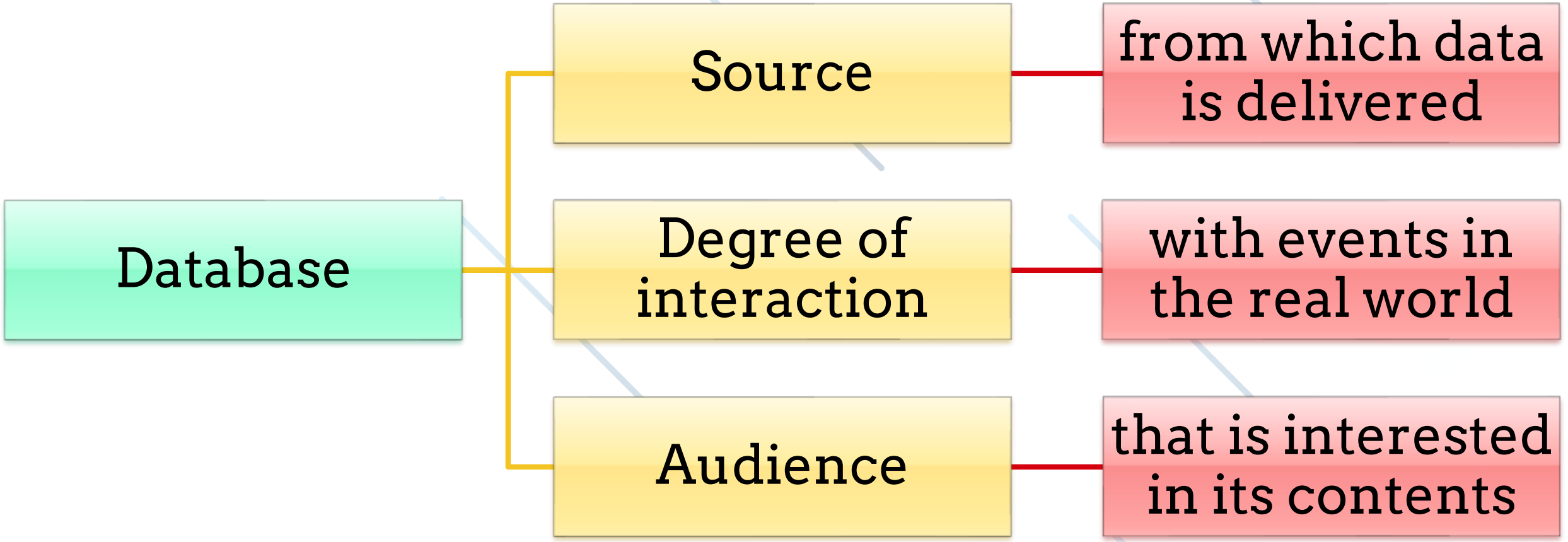
Typical DBMS Functionality

- Other features:
  - ★ ***Protection*** or Security measures to prevent unauthorized access
  - ★ ***Presentation*** and ***Visualization*** of data

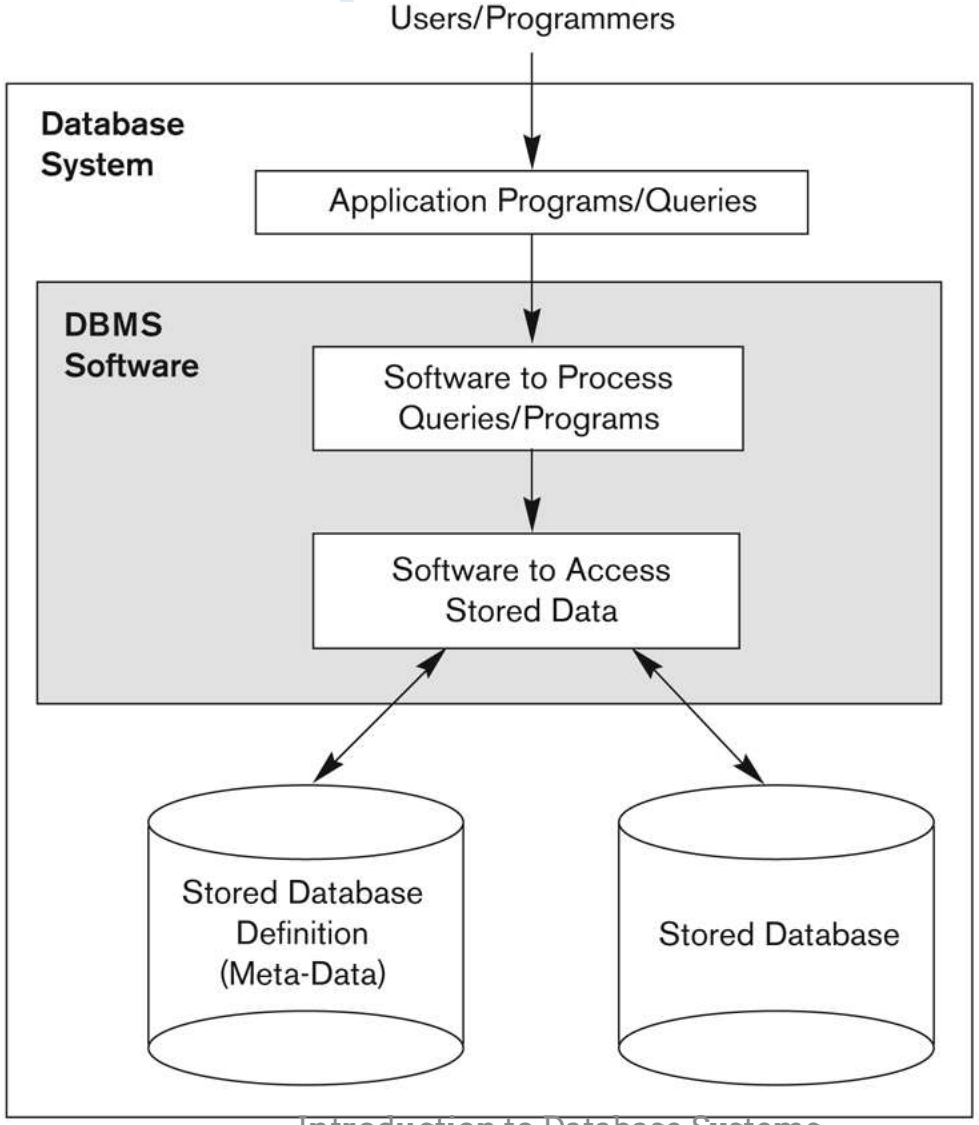
Basic Definitions (again)

- Database System:
  - The DBMS software together with the data itself. Sometimes, the applications are also included.
- Application Program:
  - Access the database by sending ***queries*** or ***requests*** for data to the DBMS.
    - ***Query*** : causes some data to be retrieved , inserted , deleted or updated.
  - Applications must not allow unauthorized users to access data.

Database Properties



Simplified database system environment



Example of a Database

- Mini-world for the example:
  - Part of a **UNIVERSITY** environment.
- Some mini-world ***entities***:
  - STUDENTs
  - COURSEs
  - SECTIONs (of COURSEs)
  - (academic) DEPARTMENTs
  - INSTRUCTORs

Example of a Database

- Some mini-world ***relationships***:
  - ★SECTIONs are of specific COURSEs
  - ★STUDENTs take SECTIONs
  - ★COURSEs have prerequisite COURSEs
  - ★INSTRUCTORs teach SECTIONs
  - ★COURSEs are offered by DEPARTMENTs
  - ★STUDENTs major in DEPARTMENTs

- Note: The above entities and relationships are typically expressed in a conceptual data model, such as the ENTITY-RELATIONSHIP data model.

Example of a simple database

Data record

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

Data elements
Each has a **data type**

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang
119	CS1310	Fall	05	Anderson
135	CS3380	Fall	05	Stone

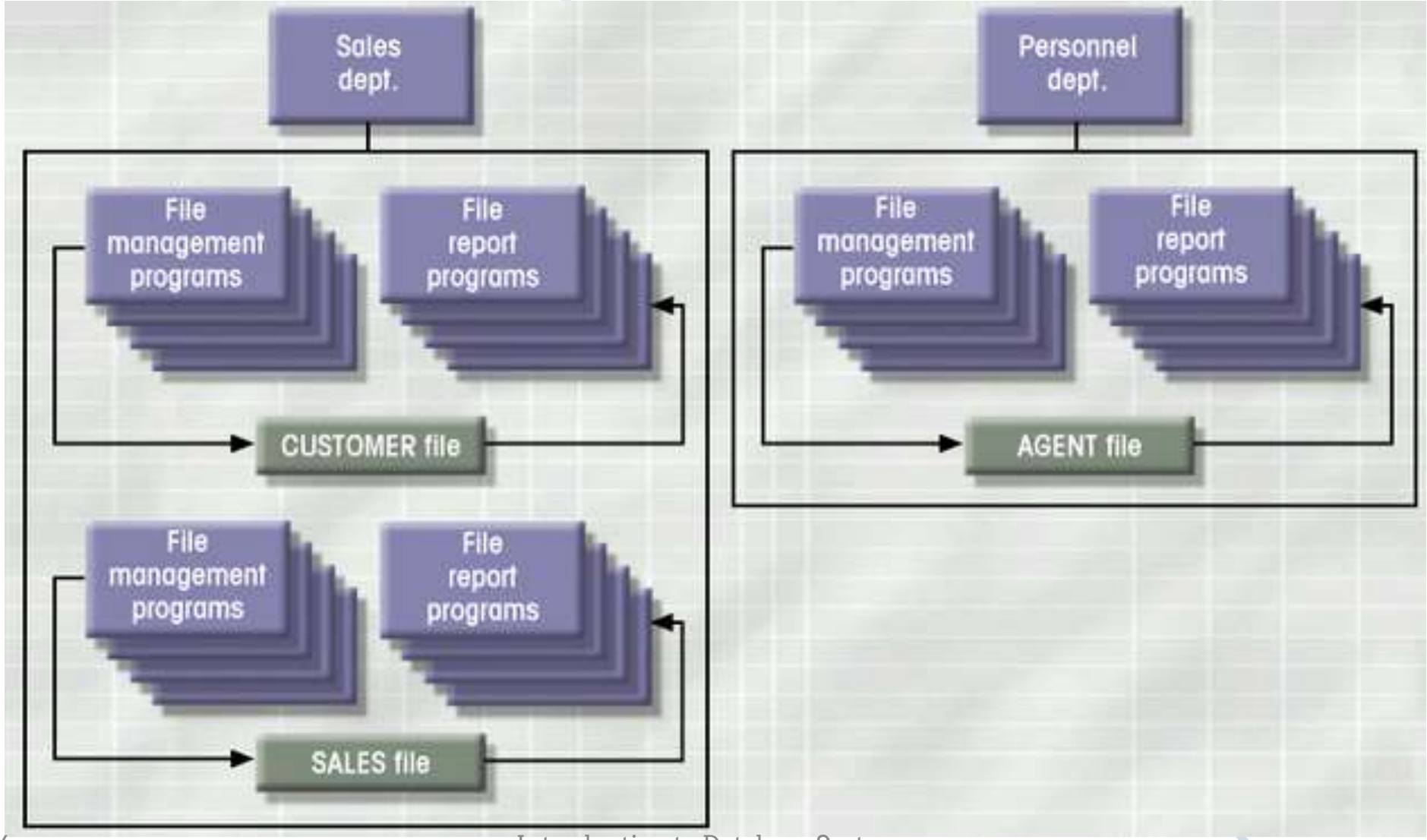
PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

File System

- Each user define and implement the files needed for a specific software application.
- Example:
  - ★Grade reporting office -> define files related to student data and grades
  - ★Accounting office : keep track of students' fees and their payments

Simple File System



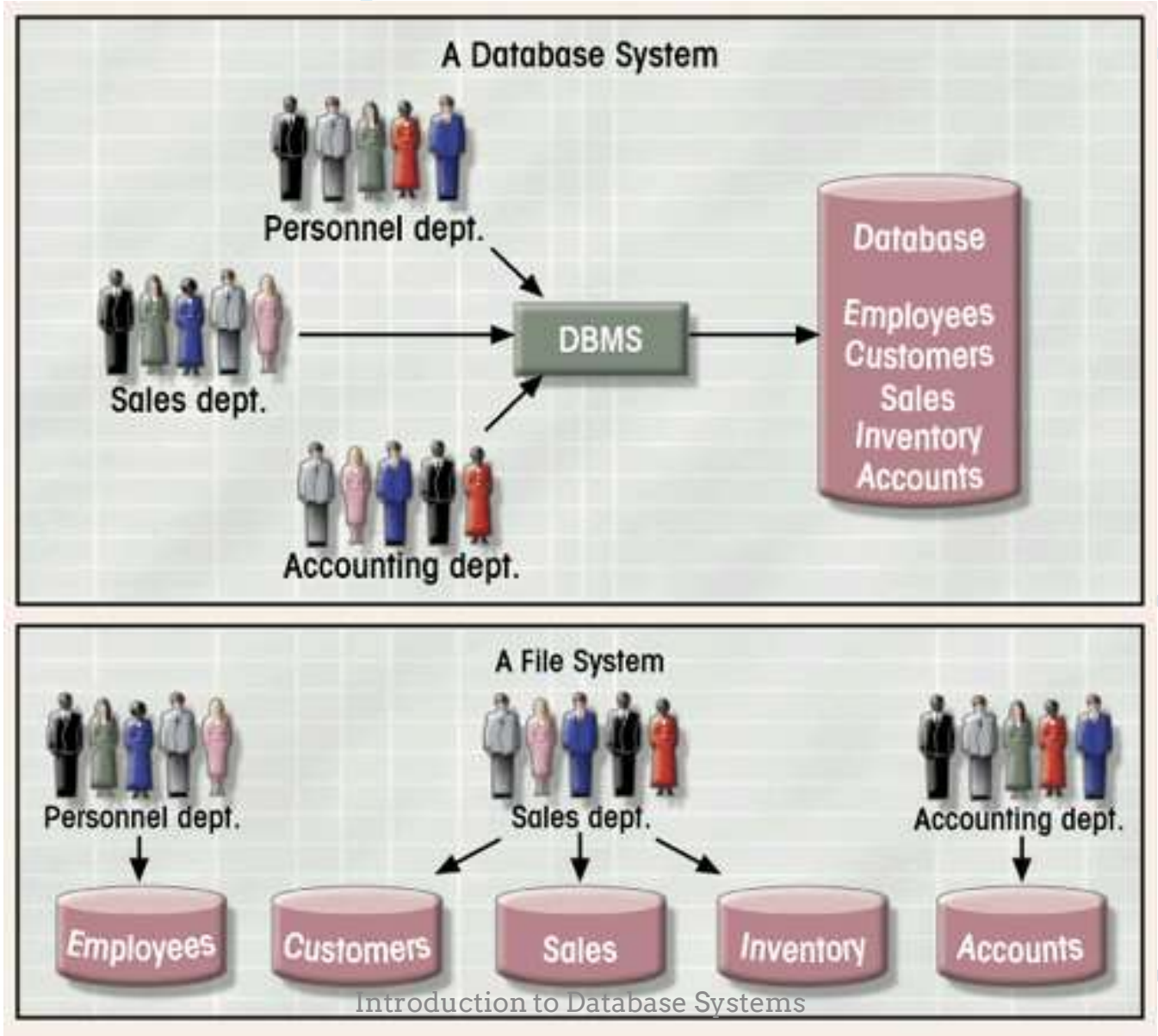
File System Critique

- Redundancy in defining and storing data which results in wasted storage space
- Redundant effort to maintain common up to date data.
- Any change to the structure of a file may require changing all programs that access that file
- Data Dependence
  - ★Change in file's data characteristics requires modification of data access programs

Database Systems

- Database consists of logically related data stored in a single repository
- Provides ***advantages over file system*** management approach
  - ★ Eliminates inconsistency, data dependency, and structural dependency problems
  - ★ Stores data structures, relationships

Database vs. File Systems



Main Characteristics of the Database Approach



Main Characteristics of DB

- Self-describing nature of a database system:
  - A ***DBMS catalog*** stores the description of a particular database (e.g. data structures, types, and constraints)
  - The information stored in the catalog is called ***meta-data***.
  - This allows the DBMS software to work with different database applications.

Example of a simplified database catalog

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
….	….	…..
….	….	…..
….	….	…..
Prerequisite_number	XXXXNNNN	PREREQUISITE

Main Characteristics of DB

- Insulation between programs and data:
  - ★ Called program-data independence.
  - ★ Allows changing data structures and storage organization without having to change the DBMS access programs.

Main Characteristics of DB

- Data Abstraction:
  - A data model is a type of data abstraction that used to hide storage details and present the users with a conceptual view of the database.
  - Programs refer to the data model constructs rather than data storage details.

Main Characteristics of DB

- Support of multiple views of the data:
  - ★ Each user may see a different view of the database, which describes only the data of interest to that user.

Example: Views

TRANSCRIPT

Student_name	Student_transcript				
	Course_number	Grade	Semester	Year	Section_id
Smith	CS1310	C	Fall	08	119
	MATH2410	B	Fall	08	112
Brown	MATH2410	A	Fall	07	85
	CS1310	A	Fall	07	92
	CS3320	B	Spring	08	102
	CS3380	A	Fall	08	135

(a)

COURSE_PREREQUISITES

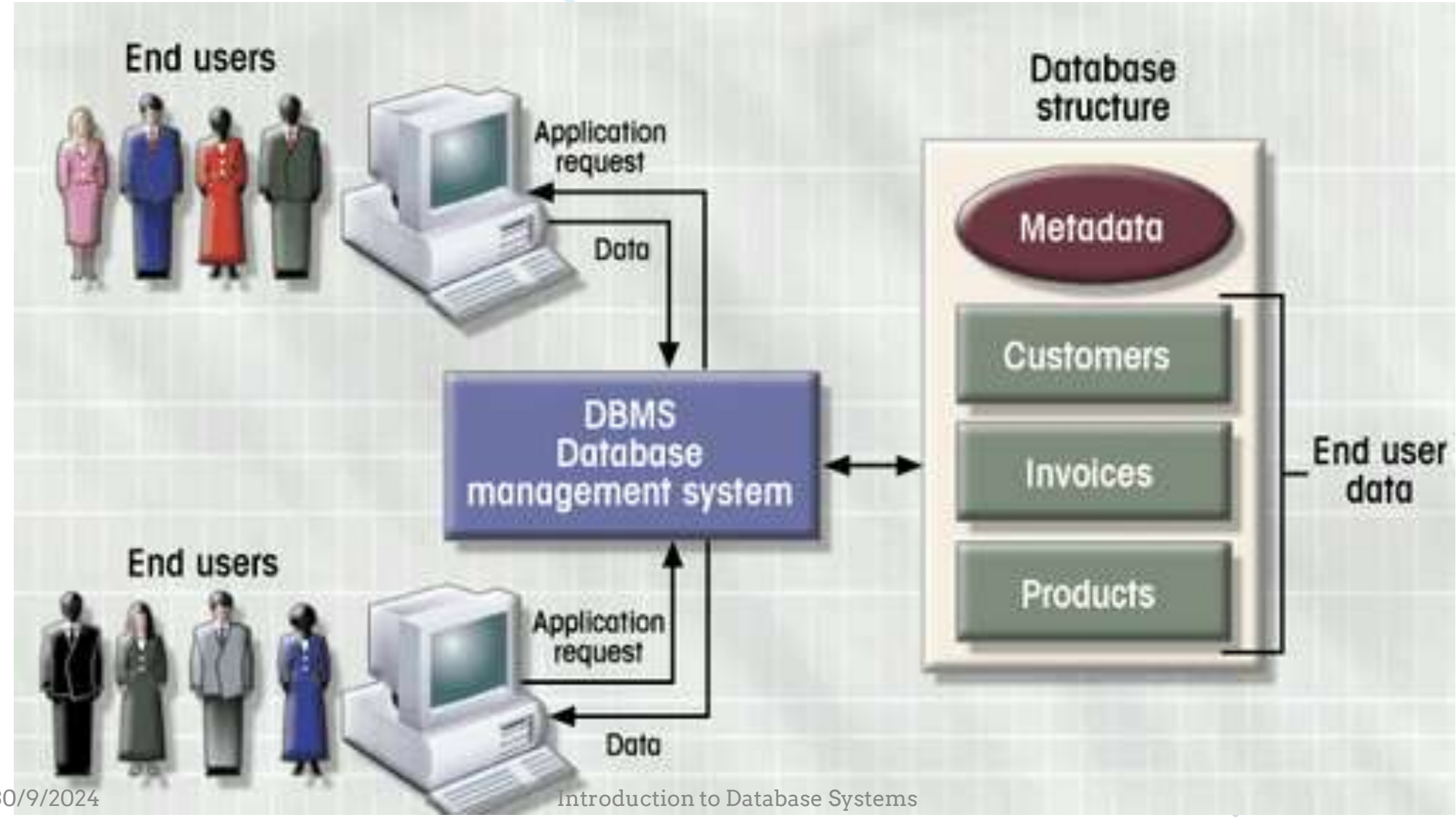
Course_name	Course_number	Prerequisites
Database	CS3380	CS3320
		MATH2410
Data Structures	CS3320	CS1310

(b)

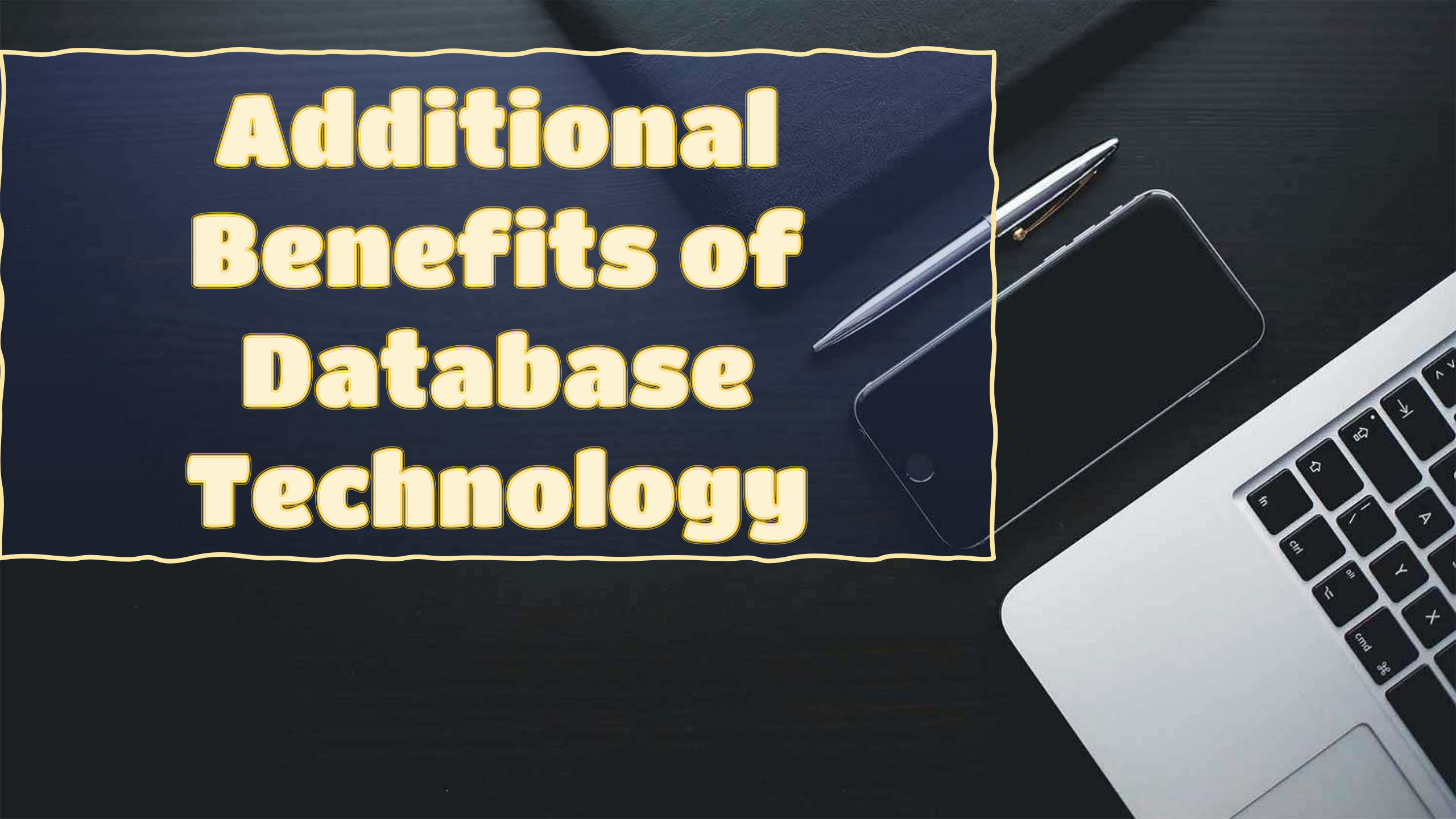
Main Characteristics of DB

- Sharing of data and multiuser transaction processing
  - ★Allow multiple users to access the database ***at the same time***.
  - ★Concurrency control within the DBMS guarantees that each transaction is ***correctly executed*** or aborted
  - ★Recovery subsystem ensures each completed transaction has its ***effect permanently recorded*** in the database.

DBMS Manages Interaction



Additional Benefits of Database Technology



Additional Benefits

- Controlling redundancy in data storage and in development and maintenance efforts.
- Sharing of data among multiple users.
- Restricting unauthorized access to data.
- Providing multiple interfaces to different classes of users.
- Representing complex relationships among data.

Additional Benefits

- Enforcing integrity constraints on the database.
- Providing backup and recovery services.
- Reduced application development time.
- Flexibility to change data structures: Database structure may evolve as new requirements are defined.
- Availability of up-to-date information.

When not to use a DBMS

- When a DBMS may be ***unnecessary***:
  - If the database and applications are simple, well defined, and not expected to change.
  - If access to data by multiple users is not required.
- When no DBMS may ***suffice***:
  - If there are stringent real-time requirements that may not be met because of DBMS overhead.
  - If the database users need special operations not supported by the DBMS (e.g., GIS and location-based services).
- When a DBMS may be ***infeasible***:
  - In embedded systems where a general-purpose DBMS may not fit in available storage.

Summary

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Thank you

Questions?

